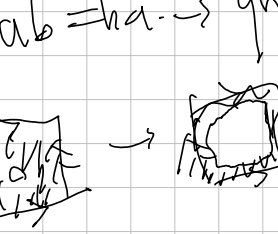
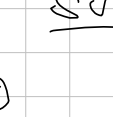


$aba^{-1}b^{-1} = e$
 $ab = ba \rightarrow \text{group } \mathbb{Z} \times \mathbb{Z}$

 $S^1 \vee S^1$


 $\frac{S^1 \vee S^1}{\mathbb{Z} \times \mathbb{Z}}$
 $T: \mathbb{Z} \times \mathbb{Z}$
 $T \cdot \mathbb{Z} \times \mathbb{Z}$
 $\mathbb{R} \cdot \mathbb{Z} \times \mathbb{Z}$
 S^1
 $S^1 \times \mathbb{R}$

[illegible][illegible]

$\mathbb{R}/\mathbb{P}^2$: 0-cell: $\{v\}$
 1-cell: $\{x, y\}$
 2-cell: $\{z\}$

$S^1 \vee S^1$

$ab^{-1}b^r = 0$
 $a, b \mid ab = b^{-1}$
 $\pi_1 \times \mathbb{Z} / a$

Handwritten notes on a grid background:

- Top left: RP
- Top center: $a^2 = 1$
- Top right: $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$
- Middle left: A diagram of a circle with a vertical line segment inside, labeled a at both ends.
- Middle center: RP followed by a list:
 - 1 cell
 - 2 cell
 - 3 cell
 - 4 cell
- Middle right: A diagram of a circle with a vertical line segment inside, labeled a at both ends.
- Bottom left: A diagram of a circle with a vertical line segment inside, labeled a at both ends.
- Bottom center: $RP = \pi(RP) = 2.7$
- Bottom right: A diagram of a circle with a vertical line segment inside, labeled a at both ends.

Hand-drawn sketches on graph paper:

- A sphere with a horizontal line passing through its center, and two points on the line.
- A flower-like shape with five petals.
- A complex loop with two dots on it.
- A spiral shape.
- A shape resembling a stylized 'S' or a wave.
- A shape resembling a stylized 'Z' or a zigzag.
- A shape resembling a stylized 'U' or a curve.
- A shape resembling a stylized 'V' or a curve.
- A shape resembling a stylized 'W' or a curve.
- A shape resembling a stylized 'X' or a curve.
- A shape resembling a stylized 'Y' or a curve.
- A shape resembling a stylized 'Z' or a curve.
- A shape resembling a stylized 'A' or a curve.
- A shape resembling a stylized 'B' or a curve.
- A shape resembling a stylized 'C' or a curve.
- A shape resembling a stylized 'D' or a curve.
- A shape resembling a stylized 'E' or a curve.
- A shape resembling a stylized 'F' or a curve.
- A shape resembling a stylized 'G' or a curve.
- A shape resembling a stylized 'H' or a curve.
- A shape resembling a stylized 'I' or a curve.
- A shape resembling a stylized 'J' or a curve.
- A shape resembling a stylized 'K' or a curve.
- A shape resembling a stylized 'L' or a curve.
- A shape resembling a stylized 'M' or a curve.
- A shape resembling a stylized 'N' or a curve.
- A shape resembling a stylized 'O' or a curve.
- A shape resembling a stylized 'P' or a curve.
- A shape resembling a stylized 'Q' or a curve.
- A shape resembling a stylized 'R' or a curve.
- A shape resembling a stylized 'S' or a curve.
- A shape resembling a stylized 'T' or a curve.
- A shape resembling a stylized 'U' or a curve.
- A shape resembling a stylized 'V' or a curve.
- A shape resembling a stylized 'W' or a curve.
- A shape resembling a stylized 'X' or a curve.
- A shape resembling a stylized 'Y' or a curve.
- A shape resembling a stylized 'Z' or a curve.

[illegible]

$$z = x + iy$$

$$K_1 = \frac{1}{4}$$

$$K_2 = \frac{1}{4}$$

$$2\pi \cdot \frac{1}{4}$$

$\int \frac{dx}{x^2} = -\frac{1}{x} + C$